

Alexander Reinthal

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7 years of industry experience - cybersecurity operations - machine learning - data science & engineering

Publications

Data Modelling for Predicting Exploits

Nov 2018

Reinthal, A, Filippakis, E., Almgren, M.

[10.1007/978-3-030-03638-6_21](https://doi.org/10.1007/978-3-030-03638-6_21) (Springer LNCS: Nordic Conference on Secure IT Systems)

Research Projects

Detecting Piecewise Cyber Espionage in Model APIs

Nov 2025 – Nov 2025

Apart Research sprint (4th place out of 641 submissions) studying detection of piecewise misuse activity in model APIs

- Showed that piecewise misuse — where individual requests appear benign — can be detected by modelling activity using the cyber kill chain
- Correlated data across requests (e.g. IP addresses) to surface patterns that per-request guardrails like Llama Guard 3 miss
- Co-authored with Arthur Colle, David Williams-King, Yingquan Li, and Lihn Le

Detecting Deception in Chinese Models

Jan 2026 – Feb 2026

ARENA 7.0 weekend hackathon (1st place) applying linear probing to detect deception and political censorship in Chinese-origin language models

- Extended "Detecting Strategic Deception Using Linear Probes" to Qwen 2.5 models, achieving AUROC 0.849 on deception detection
- Investigated whether deception and political censorship share the same linear representation in activation space using custom datasets on politically sensitive topics
- Found optimal layers for probes, conducted steering experiments, and increased deception with model size (7B to 32B), with larger models showing cleaner deception-score separation between non-sensitive and sensitive topics
- Discovered that Qwen spontaneously switches to Chinese when asked sensitive questions in English, consistent with a refusal/anti-jailbreaking mechanism

Inoculation Prompting for Emergent Misalignment

Jan 2026 – Apr 2026

ARENA 7.0 capstone project (3rd place) investigating whether inoculation prompting can prevent or reverse emergent misalignment (EM) in fine-tuned LLMs

- Inoculation (prepending "please be harmful" to training data) completely prevents EM: 0% EM across all 3 domains and encodings vs 17–34% in controls (2x5 experiment, n=2000 each)
- Discovered on-policy neutral data amplifies EM (+10pp) even when judge-screened as benign, while same questions answered by base model eliminate EM (0%), proving misalignment signal lives in EM model completions
- Identified correctness confound: EM degrades capabilities by 68pp on GSM8K, and any correct-data SFT simultaneously restores capabilities and reduces EM, making effects inseparable

Novel AI Control Protocol Classes: Evaluation and Scalability

Jan 2026 – present

Ongoing AI safety research through AI Safety Camp, investigating alternative control protocol classes and their scaling properties as model capabilities increase

- Building on Greenblatt et al.'s control evaluation framework to evaluate hierarchical and parallel control structures against simple trusted-untrusted model pairs
- Researching whether complex oversight structures offer more safety-usefulness Pareto frontiers, particularly as the capability gap between trusted and untrusted models widens

Certificates

- Bluedot Technical AI safety, Completed: 2026-03-21
- Bluedot AGI Strategy, Completed: 2025-10-07

Experience

Facilitator

Mar 2026 – Mar 2026

Bluedot Impact

- Facilitated the Technical AI safety course during a one-week intensive course. Received outstanding feedback from course participants and we collectively started a 1-month plan accountability channel on the Bluedot slack ensure we stay on target.

Alum

Jan 2026 – Feb 2026

ARENA 7.0

- Research projects placed 1st (Detecting Deception in Chinese Models) and 3rd (Inoculation Prompting Against Emergent Misalignment).
- ARENA prepares fellows for work as researchers in technical AI Safety.
- The curriculum is taught over five high-paced weeks and teaches the transformer architecture, mechanistic interpretability, RLHF, evals and working with the Inspect framework by leading researchers in the field.

Data Platform Engineer / Tech Lead / Data Scientist

Gothenburg, Sweden

Knowit Solutions Cocreate

Feb 2022 – present

- Mentored junior engineers
- Compared multiple LLM observability tools for use with a job-ad-to-consultant matching LLM platform. (LangChain, Langfuse, Arize Phoenix).
- Deployed Arize Phoenix to monitor LLM calls and model performance in production environments.

Security Analyst

Gothenburg, Sweden

NTT Security

Apr 2019 – Jan 2022

- Developed predictive analytics tooling for detecting malicious network traffic, leveraging Python, machine learning, and distributed processing.
- Provided mentorship and security training, educating analysts on threat intelligence, anomaly detection, and alert triage automation.

Python Software Engineer

Gothenburg, Sweden

Ericsson

June 2017 – Oct 2018

- Led a team to develop a log parsing and analytics tool that scaled into a dedicated engineering team at Ericsson.

Open Source and Personal Projects

Open Source Contributions

Contributions to data engineering open source projects

- Dagster Contribution: [PR #24188](#) – Enhanced workflow orchestration for ML pipeline monitoring
- DLT Hub Contribution: [PR #594](#) – Improved data ingestion reliability for ML applications

Malicious Domain Prediction System

2020 – 2021

Real-time ML prediction application for assessing malicious domains for cyber security operations

- Reduced analysis of high-volume bad domains from 1 hour per shift to less than 20 minutes per shift
- Implemented real-time inference pipeline with performance monitoring

Data Breach Analysis Pipeline

2024 – 2025

Modern data lakehouse solution for analyzing large-scale password breach data

- Developed scalable data processing pipeline using Apache Iceberg, Flink, and Spark to analyze high-volume breach data
- Passion project to test new technologies that required big data

Skills

Machine Learning & AI: PyTorch, Jax, Einops, XGBoost, SciKit Learn, Jupyter, LLM Observability (Inspect, LangChain, Langfuse, Arize Phoenix)

Programming Languages: Python, Javascript, SQL, Rust, C/C++, R, Nix, Terraform

Platforms & Tools: Modal, Runpod, Openweights, Weights & Biases, Spark, Flink, Iceberg, Databricks, Dagster, Snowflake, dbt, Apache Superset

Infrastructure & DevOps: Nix, Docker, Kubernetes, FluxCD, GitOps, CI/CD pipelines, Terraform, AWS (EC2, S3, IAM)

Education

Chalmers University of Technology

Gothenburg, Sweden

MS in Engineering Physics

Sept 2016 – June 2018

- Coursework: Statistical Physics, Neural Networks and Machine Learning

Gothenburg University

Gothenburg, Sweden

BS in Computer Science

Sept 2013 – June 2016

- Coursework: Algorithms, Testing, Debugging and Verification, Theoretical Computer Science, Operating Systems, Cryptography, Cyber Security